

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	13 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

EV Datasets (CSV Files) (Population, Specifications, Charging Stations, Global EV Sales Datasets)



Data Cleaning & Preprocessing (Python - Pandas / Excel)



Processed Dataset (CSV)



Tableau Desktop Analysis (Joins, Relationships, Calculated Fields, Filters)



Interactive Dashboard & Story (KPIs, Charts, Maps, Story Points, Filters)



End Users / Decision Makers

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for EV analysis	Tableau Desktop
2.	Application Logic-1	Data cleaning and preprocessing	Python (pandas),Excel
3.	Application Logic-2	Data transformation and relationships	Tableau Prep / Tableau Relationships
4.	Application Logic-3	Dashboard creation, filters, calculated fields, story	Tableau Desktop
5.	Database	Stores processed EV datasets	CSV Files / MySQL
6.	Cloud Database	Not used in this project	N/A
7.	File Storage	Stores datasets and Tableau workbook	Local File System (.csv, .twbx)
8.	External API-1	Not required	N/A
9.	External API-2	Not required	N/A
10.	Machine Learning Model	Not applicable	N/A
11.	Infrastructure (Server / Cloud)	Project developed and executed locally	Windows PC, Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Used Python libraries for preprocessing	Python, Pandas, NumPy
2.	Security Implementations	Local storage of datasets and workbook; restricted editing access	Windows File Permissions

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Additional EV datasets and dashboards can be added without changing the existing structure	Tableau Data Model
4.	Availability	Dashboard can be accessed anytime through Tableau Desktop or Tableau Public	Tableau Desktop/Tableau Public
5.	Performance	Optimized using cleaned datasets, relationships, filters, and calculated fields for fast visualization	Tableau Desktop